

CLINICAL AND POPULATION STUDIES

Lipoprotein(a)-Lowering by 50 mg/dL (105 nmol/L) May Be Needed to Reduce Cardiovascular Disease 20% in Secondary Prevention

A Population-Based Study

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OBJECTIVE: High Lp(a) (lipoprotein[a]) cause cardiovascular disease (CVD) in a primary prevention setting; however, it is debated whether high Lp(a) lead to recurrent CVD events. We tested the latter hypothesis and estimated the Lp(a)-lowering needed for 5 years to reduce CVD events in a secondary prevention setting.

APPROACH AND RESULTS: From the CGPS (Copenhagen General Population Study; 2003–2015) of 58 527 individuals with measurements of Lp(a), 2527 aged 20 to 79 with a history of CVD were studied. The primary end point was major adverse cardiovascular event (MACE). We also studied 1115 individuals with CVD from the CCHS (Copenhagen City Heart Study; 1991–1994) and the CIHDS (Copenhagen Ischemic Heart Disease Study; 1991–1993). During a median follow-up of 5 years (range, 0–13), 493 individuals (20%) experienced a MACE in the CGPS. MACE incidence rates per 1000 person-years were 29 (95% CI, 25–34) for individuals with Lp(a) <10 mg/dL, 35 (30–41) for 10 to 49 mg/dL, 42 (34–51) for 50 to 99 mg/dL, and 54 (42–70) for ≥100 mg/dL. Compared with individuals with Lp(a) <10 mg/dL (18 nmol/L), the multifactorially adjusted MACE incidence rate ratios were 1.28 (95% CI, 1.03–1.58) for 10 to 49 mg/dL (18–104 nmol/L), 1.44 (1.12–1.85) for 50 to 99 mg/dL (105–213 nmol/L), and 2.14 (1.57–2.92) for ≥100 mg/dL (214 nmol/L). Independent confirmation was obtained in individuals from the CCHS and CIHDS. To achieve 20% and 40% MACE risk reduction in secondary prevention, we estimated that plasma Lp(a) should be lowered by 50 mg/dL (95% CI, 27–138; 105 nmol/L [55–297]) and 99 mg/dL (95% CI, 54–273; 212 nmol/L [114–592]) for 5 years.

CONCLUSIONS: High concentrations of Lp(a) are associated with high risk of recurrent CVD in individuals from the general population. This study suggests that Lp(a)-lowering by 50 mg/dL (105 nmol/L) short-term (ie, 5 years) may reduce CVD by 20% in a secondary prevention setting.

VISUAL OVERVIEW: An online [visual overview](#) is available for this article.

Key Words: atherosclerosis ■ cardiovascular disease ■ cholesterol ■ population secondary prevention

High Lp(a) (Lipoprotein[a]) is associated with high risk of incident cardiovascular disease (CVD) in observational studies of individuals without CVD at baseline,^{1,2} that is, in a primary prevention setting. Mendelian randomization studies with genetic variants affecting the concentration of Lp(a) strongly support Lp(a) as a direct cause of incident CVD in the

form of especially coronary artery disease, but also aortic valve stenosis, heart failure, and peripheral atherosclerotic stenosis.^{3–7}

The 2016 European Guidelines for the Management of Dyslipidemias include recommendations on Lp(a) screening in high-risk individuals, including those with premature CVD, for risk classification and characterization

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Nonstandard Abbreviations and Acronyms

CCHS	Copenhagen City Heart Study
CGPS	Copenhagen General Population Study
CIHDS	Copenhagen Ischemic Heart Disease Study
CVD	cardiovascular disease
Lp(a)	lipoprotein(a)
MACE	major adverse cardiovascular event
SNP	single-nucleotide polymorphism

of dyslipidemia.⁸ The 2018 American Heart Association/American College of Cardiology Guidelines on the Management of Blood Cholesterol state that relative indications for measurement of Lp(a) are family history of premature atherosclerotic CVD or personal history of atherosclerotic CVD not explained by major risk factors, and that high Lp(a) should be considered a risk-enhancing factor.⁹ These guidelines are in line with consensus panel statements from the European Atherosclerosis Society and the National Lipid Association recommending screening for elevated Lp(a) in individuals with intermediate to high CVD risk.^{10, 11}

Despite these recommendations and the strong evidence base causally linking high Lp(a) to development of incident CVD, it is less clear whether Lp(a) is associated with risk of recurrent CVD events in individuals with preexisting CVD, that is, in a secondary prevention setting. The existing evidence is primarily based on post hoc analyses of randomized controlled trials where recruitment did not target individuals with high Lp(a) levels, where therapies were not aimed specifically at lowering Lp(a), and with significant between-study heterogeneity.^{12–15} These analyses have not surprisingly yielded mixed results, but nonetheless raised questions about the role of Lp(a) in secondary prevention^{12, 13, 16}; however, the most recent meta-analysis of placebo-controlled randomized statin outcome trials, including both primary and secondary prevention trials, showed an increased risk of CVD with increasing concentrations of Lp(a), as did post hoc analyses of the FOURIER trial.^{14, 15} Nevertheless, the available evidence on the risk of recurrent CVD events in individuals with preexisting CVD according to concentrations of Lp(a) from real-world data or outside of more selective placebo-controlled randomized clinical trials is limited.^{12, 16} Despite the fact that previous studies have addressed this question, it remains an area of controversy and strong interest.

Currently, no drugs specifically aimed at lowering Lp(a) are approved for clinical use, but an apolipoprotein(a) antisense oligonucleotide shows promising safe reductions of Lp(a) after recently ended phase 2 trials.^{17, 18} If approved, this agent would likely be used primarily in individuals with established CVD for secondary prevention

Highlights

- In this prospective cohort study of 2527 individuals from the Copenhagen General Population Study with cardiovascular disease at baseline, high concentrations of Lp(a) (lipoprotein[a]) were associated with high risk of recurrent cardiovascular disease.
- Lp(a)-lowering by 50 mg/dL (105 nmol/L) for 5 years may be needed to reduce cardiovascular disease by 20% in a secondary prevention setting.
- There is a possible unmet need for secondary prevention in individuals with increased lipoprotein(a), and they could be a target group for future randomized cardiovascular outcome trials.

of recurrent CVD events, or in individuals with familial hypercholesterolemia.¹⁹

We hypothesized that high concentrations of Lp(a) are associated with high risk of recurrent CVD events in individuals from the general population with preexisting CVD. We also estimated the Lp(a) lowering needed for 5 years to reduce CVD events in a secondary prevention setting.

METHODS

The data that support the findings of this study are available from the corresponding author on reasonable request. A detailed description of the Methods is given in the online-only Data Supplement.

Study Cohorts

The primary study cohort was comprised of individuals from the CGPS (Copenhagen General Population Study), in which white adults of Danish descent were included between 2003 and 2015. A total of 58 527 individuals from the CGPS had baseline measurements of Lp(a). For this study, individuals aged 80 or older ($n=2691$) and all individuals without CVD at baseline ($n=53 309$) were excluded, leaving a total of 2527 individuals with CVD defined as previous myocardial infarction, coronary artery bypass graft, percutaneous coronary intervention, ischemic stroke, or peripheral arterial disease of the lower extremities.

For confirmation of the results, individuals from 2 other cohorts, the CCHS (Copenhagen City Heart Study) and the CIHDS (Copenhagen Ischemic Heart Disease Study), were also included and combined. In the CCHS, individuals were recruited and examined similarly to those in the CGPS. CVD at baseline was defined the same way as in the CGPS. From the 1991 to 1994 examination, 321 individuals below age 80 and with CVD at baseline were included. The CIHDS included patients from the greater Copenhagen area referred for coronary angiography to Copenhagen University Hospital. For the present study, 794 patients admitted during the period 1991 to 1993 with baseline measurement of Lp(a) and angiographic evidence of coronary disease (severe stenosis, ie, >70% stenosis of at least one coronary artery or >50% stenosis of the left main artery) were included.

Laboratory Analyses

Blood samples were nonfasting as recommended by the European Atherosclerosis Society and European Federation of Clinical Chemistry and Laboratory Medicine.^{20, 21} Lp(a) total mass (mg/dL) was measured using apolipoprotein(a) turbidimetric assays (Table I in the [online-only Data Supplement](#)).^{1, 22} Of the 2527 individuals included from the CGPS, the first 245 had Lp(a) measured immediately after sampling using an assay from DiaSys (Diagnostic Systems). The remaining individuals in the CGPS had Lp(a) measurements done using an assay from Denka Seiken in mg/dL. The assay uses a polyclonal antibody but displays minimal isoform bias as it is latex based and uses 5 independent calibrators with suitable distributions of apo(a) isoforms and with values covering the concentration spectrum from high to low. The calibrators are manufactured to reflect clinical status to the maximum by selecting donors with required Lp(a) levels. Values in mg/dL are assigned using dedicated master calibrators with traceability controlled by an in-house standard (Denka Seiken). Of the individuals with measurements done using the Denka Seiken assay in mg/dL, samples from 520 individuals were stored at -80°C until measurement, and 1762 were measured on fresh samples. All individuals from the CCHS and the CIHDS had Lp(a) measurements performed shortly after sampling using an in-house assay.¹ To avoid bias from the use of different assays, all measurements of Lp(a) were uniformly calibrated to the concentrations from the Denka Seiken assay in mg/dL on fresh samples to reflect a commercially available assay, as done previously.²³ For comparison, the Lp(a) concentrations calibrated to the concentrations from the Denka Seiken assay in mg/dL on fresh samples were converted to nmol/L using the following equation: $\text{Lp(a), nmol/L} = 2.18 \times \text{Lp(a), mg/dL} - 3.83$ (Figure I in the [online-only Data Supplement](#)). The negative constant bias of 3.83 means the molar to mass ratio of Lp(a) varies with the concentration of Lp(a). The equation is based on a linear regression of 13930 individuals from the CGPS with measurements of Lp(a) in fresh samples using both the mass assay in mg/dL and the molar assay in nmol/L by Denka Seiken distributed by Roche with high correlation (R^2 of 0.996) on the same sample, as described previously.²⁴ The 13930 individuals are not the same individuals as included in the present study. The assay in nmol/L is standardized against SRM-2B, the WHO approved Lp(a) reference material. The master calibrators used for value assignment for the mg/dL and the nmol/L assays are different but highly correlated.

The kringle IV type 2 number of repeats were measured by real-time polymerase chain reaction. The assay was designed to determine the number of copies of kringle IV type 2 which is part of the apolipoprotein(a) gene and the assay determines the sum of copies on both alleles.³ A total of 2131 individuals in the CGPS had measurements of kringle IV type 2 number of repeats. The 2 single-nucleotide polymorphisms (SNPs) *LPA* rs10455872 and rs3798220 were genotyped using Taqman assays and were available in 2520 individuals from the CGPS.

Total and high-density lipoprotein cholesterol and plasma triglycerides were measured using colorimetric assays (Konelab). LDL cholesterol was calculated using the Friedewald equation when triglycerides were below 4 mmol/L (354 mg/dL) or otherwise measured directly (Konelab). Conventional LDL cholesterol assays measure the cholesterol content in both LDL

and Lp(a) particles, and 30% to 45% of the mass of Lp(a) is cholesterol.²⁵ Lp(a)-corrected measurements of LDL cholesterol used for adjustment and stratification were therefore calculated using the conservative estimate of 30% by subtracting the cholesterol content in Lp(a) from the measured LDL cholesterol.²⁵

End Point

The primary end point was the composite end point major adverse cardiovascular event (MACE). The end point was constructed to mimic those often used in cardiovascular randomized placebo-controlled trials and included first occurrence of cardiovascular death (*International Classification of Diseases [ICD-10]*: I01-I87, I95-I99, R96), nonfatal myocardial infarction (*ICD-10*: I21-I22), coronary artery bypass graft (NOMESCO: KNFA-KFNE), percutaneous coronary intervention (NOMESCO: KFNG00-05), and stroke (*ICD-10*: I60-I61, I63-I64) following baseline.

Statistical Analyses

Lp(a) were divided into 4 categories taking into consideration statistical power and exploration of an extreme elevated Lp(a) phenotype. Absolute risks of MACE were given as incidence rates per 1000 person-years with 95% CIs calculated based on the Poisson distribution, and as cumulative incidence curves estimated nonparametrically using the cumulative incidence function, with death (excluding cardiovascular deaths) and emigration ($n=4$) as competing events. Cumulative incidences were compared using Gray's test. The risk of MACE was further analyzed using three different types of rate ratios: (1) incidence rate ratios, (2) hazard ratios, and (3) subhazard ratios. Incidence rate ratios were estimated using Poisson regression. Hazard ratios were estimated using Cox proportional hazards regression and subhazard ratios using Fine and Gray competing risk regression with time on study as the underlying timescale in unadjusted analyses and age as the underlying timescale in multifactorially adjusted analyses. For multifactorially adjusted Cox proportional hazards regression, potential confounders were chosen a priori as covariates possibly associated with Lp(a) and MACE, and included age, sex, lipid-lowering therapy ($>98\%$ statins), diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, LDL cholesterol, and triglycerides. Missing covariates were imputed (Tables II and III in the [online-only Data Supplement](#)). *P* values for interactions were obtained using the likelihood-ratio test.

Estimated Lp(a)-lowering required for 5 years in a secondary prevention setting, for a given MACE risk reduction based on plasma Lp(a) was calculated using estimates from a multifactorially adjusted Cox proportional hazards regression with Lp(a) as a continuous variable. The functional form of the Lp(a) recurrent MACE event relationship was assessed using splines (Figure II in the [online-only Data Supplement](#)) and by formal testing incorporating splines, quadratic, reciprocal, or logarithmic terms all with $P>0.05$. Hence, we did not reject the linear form of untransformed Lp(a) as the best-fitting model. Based on this, a MACE risk reduction per 10 mg/dL reduction in Lp(a) was obtained and converted to the reduction in Lp(a) needed for a given reduction in MACE risk as: $[10 \text{ mg/dL reduction in Lp(a)}] \times [\text{desired reduction in risk of MACE (eg,$

20%)]/[1-hazard ratio for MACE per 10 mg/dL reduction in Lp(a)]. 95% CIs were obtained from bootstrapping. These estimates were compared with previously reported estimates in primary prevention settings based on SNPs for Lp(a) and with the 22% major vascular event risk reduction per 1 mmol/L (39 mg/dL) reduction in LDL cholesterol observed in combined primary and secondary prevention statin outcome trials from the Cholesterol Treatment Trialists' Collaboration meta-analysis.²⁶ The median follow-up of 5 years in the Cholesterol Treatment Trialists' Collaboration meta-analysis was similar to our study. Estimated Lp(a)-lowering required for a given MACE risk reduction was also estimated based on genetically predicted Lp(a). This analysis was based on kringle IV type 2 number of repeats as a continuous variable and the 2 *LPA* SNPs. For the main analysis, we performed instrumental variable analysis using the 2-sample predictor substitution method with a first-stage linear regression of Lp(a) on kringle IV type 2 and the 2 SNPs adjusted for age and sex and a second-stage logistic regression including the predictors from the first stage and adjusted for age and sex. The obtained instrumental variable estimates reflect a lifelong lower concentration of Lp(a). To scale this to a short-term trial setting (ie, 5 years), we assumed, similar to the previously reported estimates from primary prevention,^{27,28} that the fraction between lifelong and short-term risk reduction is similar for LDL cholesterol and Lp(a). Here, we used the expected 54% reduction in risk of coronary heart disease per 1 mmol/L (38.7 mg/dL) per lowering of genetically predicted LDL cholesterol from meta-analyses of previous mendelian randomization studies.²⁹ This was compared with the short-term reduction of major coronary event risk of 24% per 1 mmol/L (38.7 mg/dL) lower LDL cholesterol observed in a

meta-analysis of statin trials from the Cholesterol Treatment Trialists' Consortium from 2010 including both primary prevention and secondary prevention trials.²⁶ 95% CIs for the estimated short-term effect of Lp(a)-lowering were obtained from bootstrapping.

RESULTS

The concentrations of Lp(a) were higher in the 2527 individuals from the CGPS aged 20 to 79 with CVD at baseline as compared with the 53 309 individuals aged 20 to 79 without (Figure III in the online-only Data Supplement). At baseline, coronary artery disease (1445 [57%]) was the most common CVD followed by peripheral arterial disease (690 [27%]) and ischemic stroke (610 [24%]). The median time since first diagnosis of CVD to baseline was 4 years (range, 0–34). During a median follow-up of 5 years (range, 0–13, 13 974 person-years), 493 individuals (20%) experienced a MACE resulting in an overall incidence rate of 35 (95% CI, 32–39) per 1000 person-years. Baseline characteristics according to concentrations of Lp(a) are shown in the Table.

Risk of MACE According to Concentrations of Lp(a)

The absolute risk of MACE increased with stepwise increasing concentrations of Lp(a) (Figure 1). Individuals

Table. Baseline Characteristics in the Copenhagen General Population Study According to Concentrations of Lp(a)

	mg/dL	Lp(a)				P Value
		<10	10–49	50–99	≥100	
	nmol/L	<18	18–104	105–213	≥214	
	Percentiles	1–43	44–75	76–92	>92	
Individuals		1086	792	437	212	
Men		727 (67)	503 (64)	284 (65)	113 (53)	0.002
Age, y		68 (62–73)	69 (61–75)	69 (63–74)	68 (62–74)	0.37
Lipid-lowering therapy		709 (65)	504 (64)	311 (71)	175 (83)	<0.001
Diabetes mellitus		180 (17)	101 (13)	65 (15)	37 (17)	0.11
Hypertension		906 (83)	662 (84)	370 (85)	177 (83)	0.95
Body mass index, kg/m ²		27.4 (25.0–30.4)	26.9 (24.6–30.0)	27.3 (24.7–30.3)	27.7 (24.9–30.6)	0.15
Current smoker		219 (20)	161 (20)	96 (22)	41 (19)	0.85
Alcohol intake, units/wk*		9 (4–17)	10 (4–16)	10 (4–19)	9 (4–17)	0.88
Physical inactivity†		606 (56)	418 (53)	244 (56)	119 (56)	0.56
LDL cholesterol						
mmol/L		2.4 (1.8–3.1)	2.4 (1.9–3.3)	2.4 (1.9–3.1)	2.6 (2.1–3.1)	0.006
mg/dL		93 (70–120)	93 (73–127)	94 (74–121)	100 (82–121)	0.006
Nonfasting triglycerides						
mmol/L		1.7 (1.1–2.4)	1.6 (1.1–2.3)	1.5 (1.1–2.3)	1.6 (1.1–2.4)	0.10
mg/dL		147 (98–216)	143 (95–200)	135 (94–204)	145 (101–209)	0.10

Values are median (interquartile range) or number of individuals (%). P value are from Pearson χ^2 test for categorical variables and Kruskal-Wallis test for continuous variables. The number of individuals varies slightly according to availability of the variables (data are without imputation). Lp(a) indicates lipoprotein(a).

*1 unit $\approx$ 12 g of alcohol.

†Less than 4 h physical activity in leisure time per week.

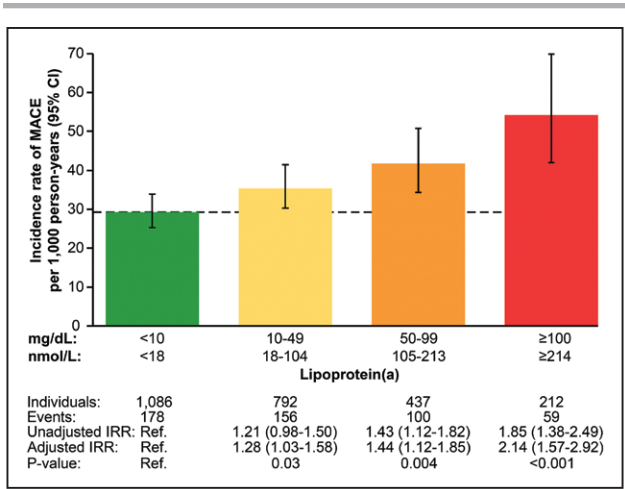


Figure 1. Absolute risk of major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]).

Based on 2527 individuals from the Copenhagen General Population Study. Incidence rates and incidence rate ratios with 95% CIs are from Poisson regression. Error bars indicate 95% CIs. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides. IRR indicates incidence rate ratio.

with Lp(a)<10 mg/dL (<18 nmol/L) had the lowest incidence rate of 29 (95% CI, 25–34) per 1000 person-years for MACE, and the incidence rate for individuals with Lp(a) of 10 to 49 mg/dL (18–104 nmol/L) was 35 (30–41), for 50 to 99 mg/dL (105–213 nmol/L) 42 (34–51), and for ≥100 mg/dL (≥214 nmol/L) 54 (42–70). In individuals with Lp(a)<10 mg/dL as the reference, the respective multifactorially adjusted incidence rate

ratios were 1.28 (1.03–1.58) for individuals with Lp(a) of 10 to 49 mg/dL (18–104 nmol/L), 1.44 (1.12–1.85) for 50 to 99 mg/dL (105–213 nmol/L), and 2.14 (1.57–2.92) for ≥100 mg/dL (≥214 nmol/L). Similar results were obtained from analyses based on percentiles from the individual assays, as opposed to the uniformly calibrated absolute concentrations of Lp(a) (Figure IV in the [online-only Data Supplement](#)).

The cumulative incidence of MACE increased with higher concentrations of Lp(a) with 5-year risks of MACE of 13.2% (11.1–15.4) for individuals with Lp(a) of <10 mg/dL (<10 nmol/L), 15.6% (13.0–18.3) for 10 to 49 mg/dL (18–104 nmol/L), 17.2% (13.7–21.1) for 50 to 99 mg/dL (105–213 nmol/L), and 19.3% (14.1–25.2) for ≥100 mg/dL (≥214 nmol/L; Figure 2). As compared with individuals with Lp(a)<10 mg/dL, the multifactorially adjusted subhazard ratios were 1.29 (1.04–1.59) for individuals with Lp(a) of 10 to 49 mg/dL (18–104 nmol/L), 1.46 (1.14–1.89) for 50 to 99 mg/dL (105–213 nmol/L), and 2.17 (1.59–2.98) for ≥100 mg/dL (≥214 nmol/L). The results were similar in unadjusted Cox proportional hazards regression, and multifactorial adjustment only changed the risk estimates slightly (Figure 3). In comparison, the corresponding hazard ratios for MACE in the 53 309 individuals without CVD at baseline were slightly smaller (Figure V in the [online-only Data Supplement](#)).

Stratified Analyses

When comparing individuals with Lp(a)≥50 mg/dL (≥105 nmol/L) to individuals with Lp(a)<10 mg/dL

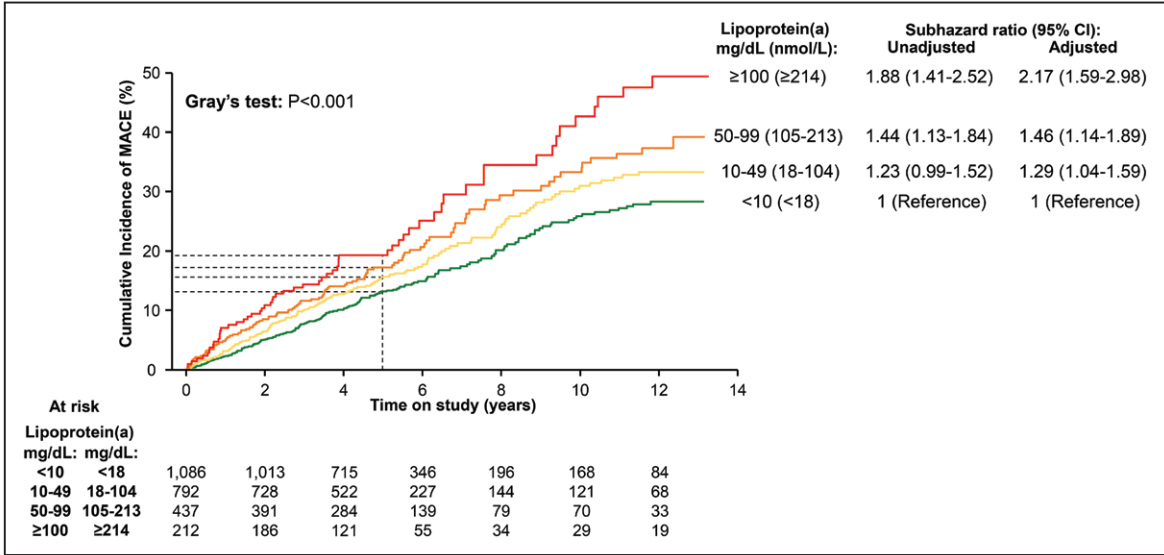


Figure 2. Cumulative incidence of major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]).

Based on 2527 individuals from the Copenhagen General Population Study. Cumulative incidence curves were estimated with death (excluding cardiovascular deaths for the MACE end point) and emigration as competing events, and with time on study as the underlying timescale. Subhazard ratios were from Fine and Gray competing risk regression. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides.

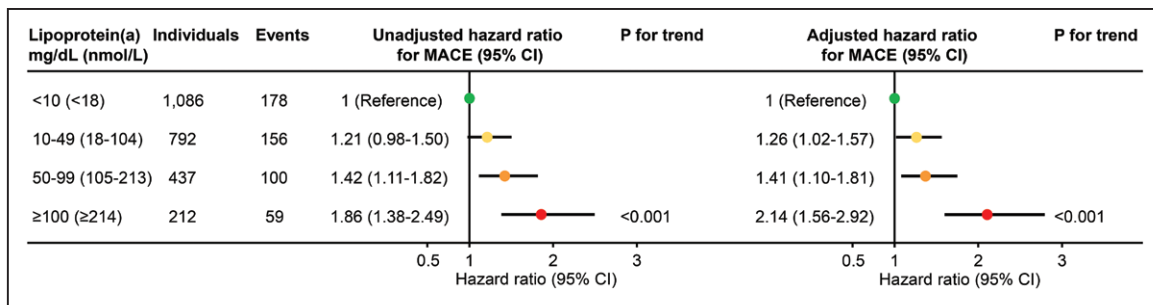


Figure 3. Hazard ratios for major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]).

Based on 2527 individuals from the Copenhagen General Population Study. Hazard ratios and 95% CIs are from Cox proportional hazards regression. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides.

(<18 nmol/L), the absolute differences in incidence rates and the incidence rate ratios for MACE were similar across different strata of covariates, except for analyses stratified by age (*P* value for interaction 0.004, all other >0.05). Thus, individuals below age 60 had a higher absolute increase in incidence rate of 37 and a higher incidence rate ratio for MACE of 3.42 (2.02–5.80), than older individuals (Figure 4). Specifically, in analyses

stratified for Lp(a)-corrected LDL cholesterol, the incidence rates of MACE in individuals with Lp(a) <10 mg/dL (<18 nmol/L) and ≥50 mg/dL (≥105 nmol/L) were 29 (25–34) and 46 (39–53) in all individuals, and similar in individuals with Lp(a)-corrected LDL-cholesterol <2.6 mmol/L (100 mg/dL) or <1.8 mmol/L (70 mg/dL), that is, 29 (24–35) and 45 (38–54), and 31 (23–41) and 46 (37–57), respectively (Figure 5). Also, when stratifying

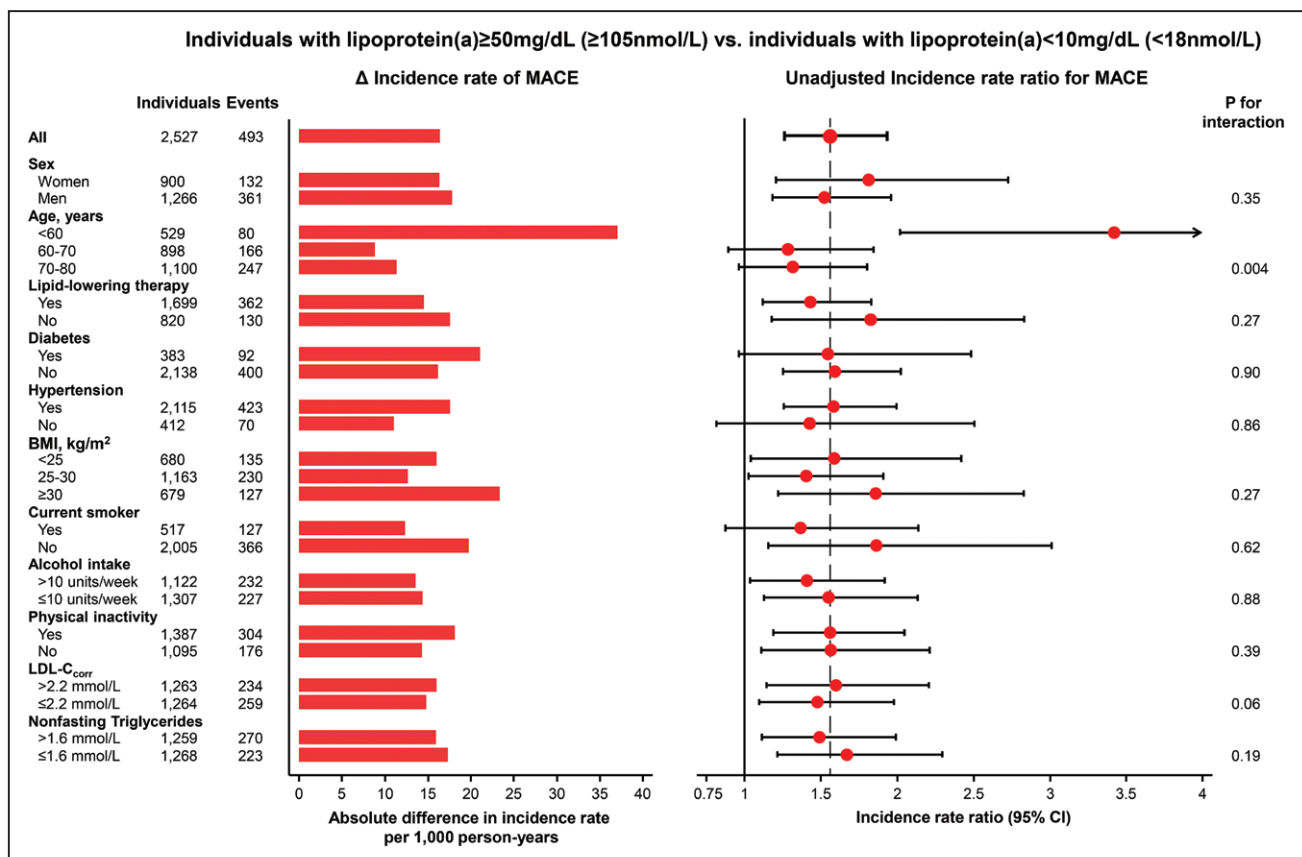


Figure 4. Risk of major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]) in different strata.

Based on 2527 individuals from the Copenhagen General Population Study. Incidence rate ratios with 95% CIs are from Poisson regression. *P* value for interaction was from the likelihood test following incorporation of 2-factor interaction terms in the full model with Lp(a) on a continuous scale, including all individuals, and not only individuals with Lp(a) ≥50 mg/dL vs individuals with Lp(a) <10 mg/dL, as shown here for simplicity. LDL cholesterol was corrected for Lp(a). Alcohol intake, LDL-C_{corr}, and nonfasting triglycerides were grouped according to the 50th percentile. BMI indicates body mass index; IRR, incidence rate ratio; and LDL-C_{corr}, Lp(a)-corrected low-density lipoprotein cholesterol.

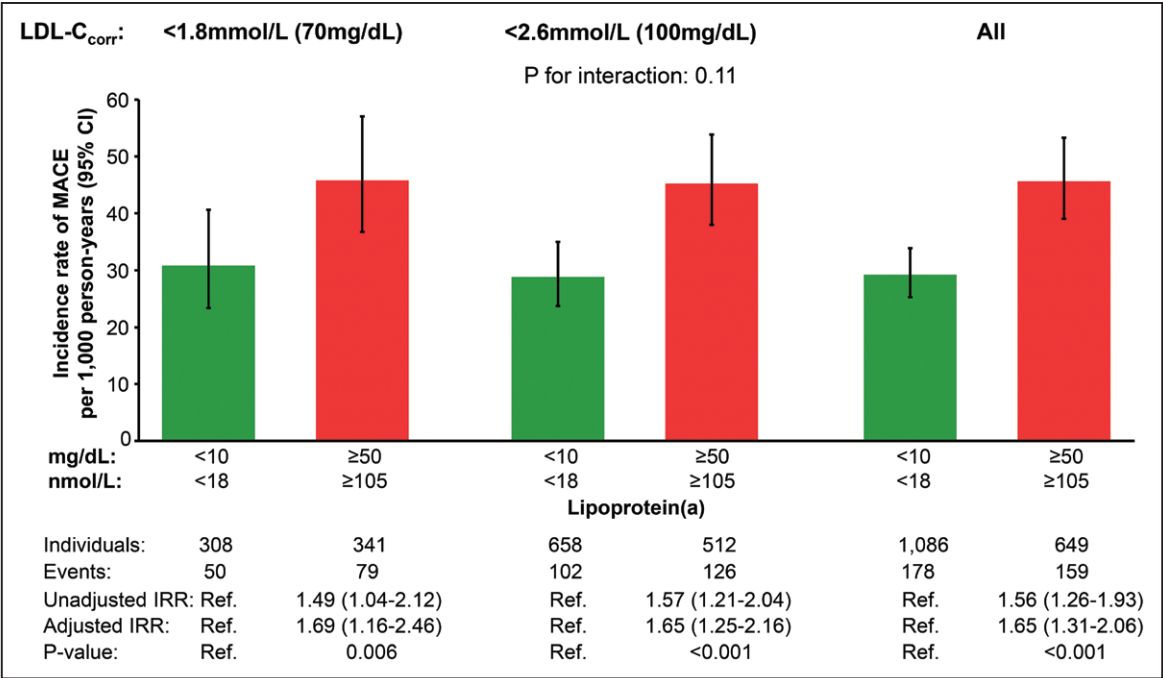


Figure 5. Absolute risk of major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]) stratified by LDL cholesterol.

Based on 2527 individuals from the Copenhagen General Population Study. Incidence rates and incidence rate ratios with 95% CIs are from Poisson regression. Error bars indicate 95% CIs. LDL cholesterol was corrected for the cholesterol contribution from Lp(a). *P* value for interaction was from the likelihood-ratio test following incorporation of 2-factor interaction terms in the full model with Lp(a) on a continuous scale, including all individuals, and not only individuals with Lp(a)≥50 mg/dL vs individuals with Lp(a)<10 mg/dL, as shown here for simplicity. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides. IRR indicates incidence rate ratio; and LDL-C_{corr}, Lp(a)-corrected low-density lipoprotein cholesterol.

according to time since the first qualifying ASCVD event, the association between high Lp(a) and high risk of MACE seemed most pronounced in those with >2 years since the first qualifying CVD event to baseline; however, there was no statistically significant interaction with time since qualifying ASCVD event (Figure VI in the [online-only Data Supplement](#)).

Risk of MACE According to Genetically Predicted Lp(a)

In genetic analyses, decreasing number of kringle IV type 2 repeats and hence increasing concentrations of Lp(a) were associated with increasing risk of MACE (Figure 6). Similarly, carriers of Lp(a)-increasing alleles had higher risk of MACE compared with noncarriers. A 10 mg/dL (18 nmol/L) lower genetically predicted Lp(a) was associated with a causal risk ratio for MACE of 0.93 (95% CI, 0.89–0.96) when including kringle IV type 2 repeats, and both SNPs in an instrumental variable analysis. Results were similar when using a Cox regression model for the second stage regression (Table IV in the [online-only Data Supplement](#)). Baseline characteristics according to the 3 instruments are shown in Tables V through VII in the [online-only Data Supplement](#).

Estimated Lp(a)-Lowering Required for MACE Risk Reduction in Secondary Prevention

A 10 mg/dL (18 nmol/L) lower plasma Lp(a) was associated with a multifactorially adjusted hazard ratio for MACE of 0.96 (95% CI, 0.94–0.98) during 5 years of follow-up. Based on this, an estimated absolute reduction in Lp(a) of 50 mg/dL (95% CI, 27–138; 105 nmol/L [55–297]) and 99 mg/dL (54–273; 212 nmol/L [114–592]) would be needed to obtain reductions in risk of MACE of 20% and 40% over a median follow-up of 5 years (Figure 7).

In the 2010 Cholesterol Treatment Trialists' Collaboration meta-analysis, it was estimated that the effect of lowering LDL cholesterol 1 mmol/L (39 mg/dL) is a 22% reduction in risk of major vascular events across primary and secondary prevention trials, also with a median follow-up of 5 years. In the present study, an absolute reduction in Lp(a) total mass of 55 mg/dL (116 nmol/L) and of Lp(a) cholesterol content of 0.4 mmol/L (17 mg/dL) would be needed to obtain the same 22% reduction as observed for lowering LDL cholesterol 1 mmol/L (39 mg/dL); we here assume that 30% of the total mass of Lp(a) is cholesterol, as done previously.¹⁹

The genetic estimate representing a 10 mg/dL (18 nmol/L) lower lifelong exposure to Lp(a) was scaled to

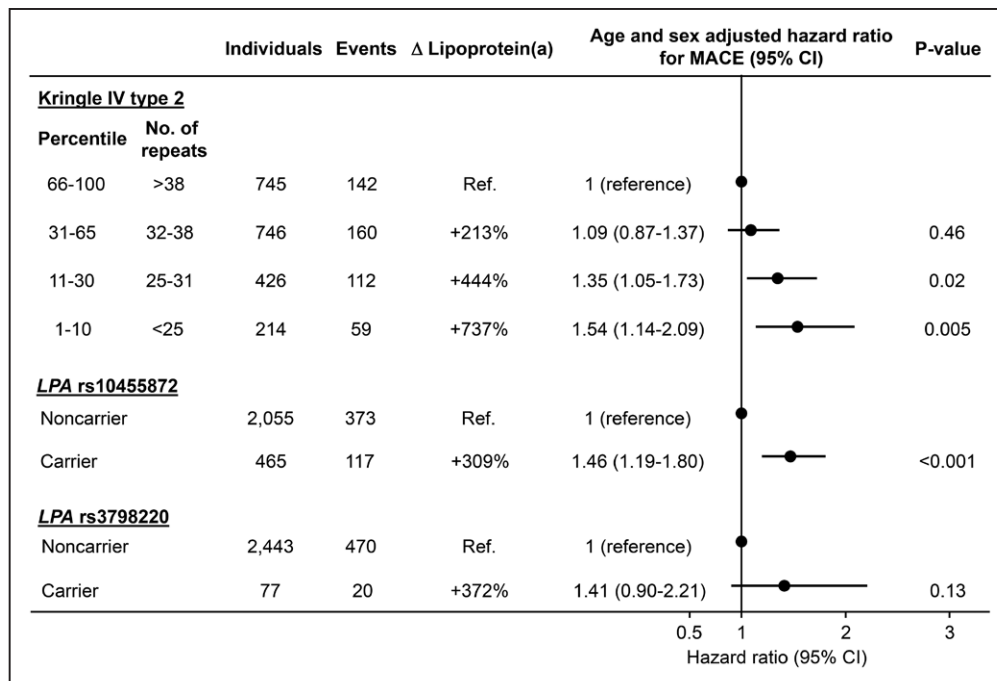


Figure 6. Hazard ratios for major adverse cardiovascular event (MACE) according to kringle IV type 2 repeats and LPA single-nucleotide polymorphisms.

Based on 2131 to 2520 individuals from the Copenhagen General Population Study. Hazard ratios and 95% CIs are from age- and sex-adjusted Cox proportional hazards regression.

an estimated risk reduction in a short-term clinical trial setting. Based on this, a genetically estimated short-term (ie, 5 years) absolute reduction in Lp(a) of 60 mg/dL (95% CI, 42–109; 127 nmol/L [88–234]) and 120 mg/dL (85–225; 258 nmol/L [181–487]) would be needed to obtain reductions in risk of MACE of 20% and 40% (Figure 7).

Independent Confirmation—The CCHS and The CIHDS

A total of 321 individuals with CVD at baseline were included from the CCHS and 794 individuals from the CIHDS. The median follow-up was 4 years (range, 0–25, 7432 person-years) and 840 (75%) experienced a MACE giving an overall incidence rate of 113 (95% CI, 106–121) per 1000 person-years, which was 3.2-fold as high as the overall incidence rate observed in the CGPS. Baseline characteristics according to concentrations of Lp(a) in the CCHS and the CIHDS are shown in Tables VIII and IX in the [online-only Data Supplement](#).

As in the CGPS, there was a stepwise increasing risk of MACE with increasing concentrations of Lp(a) (Figure 8). The incidence rates per 1000 person-years were 94 (95% CI, 84–106) for individuals with Lp(a) <10 mg/dL (<18 nmol/L), 115 (103–129) for 10 to 49 mg/dL (18–104 nmol/L), 134 (115–156) for 50 to 99 mg/dL (105–213 nmol/L), and 140 (116–169) for ≥100

mg/dL (≥214 nmol/L). Compared with individuals with Lp(a) <10 mg/dL (<18 nmol/L), the multifactorially adjusted incidence rate ratio for MACE for individuals with Lp(a) of 10 to 49 mg/dL (18–104 nmol/L) was 1.22 (1.03–1.44), for 10 to 49 mg/dL (18–104 nmol/L) 1.50 (1.23–1.83), and for ≥100 mg/dL (≥214 nmol/L) 1.65 (1.31–2.07).

DISCUSSION

In individuals from the general population with preexisting CVD, this study found that high concentrations of Lp(a) are associated with high risk of recurrent CVD. This study suggests that Lp(a)-lowering by 50 mg/dL (105 nmol/L) short-term (ie, 5 years) may reduce CVD by 20% in a secondary prevention setting. For direct comparison with lowering LDL cholesterol 1 mmol/L (39 mg/dL) causing a 22% reduction in risk of major vascular events, the corresponding needed reduction in secondary prevention in Lp(a) total mass is 55 mg/dL (116 nmol/L) and in Lp(a) cholesterol content 0.4 mmol/L (17 mg/dL).

Available evidence from mechanistic, observational, and genetic studies support a causal role of Lp(a) in the development of different types of CVD.^{30,31} However, the majority of human evidence on the role of Lp(a) in cardiovascular health stems from studies focusing on the risk of incident CVD in a primary prevention setting, and not recurring events in individuals with established CVD in

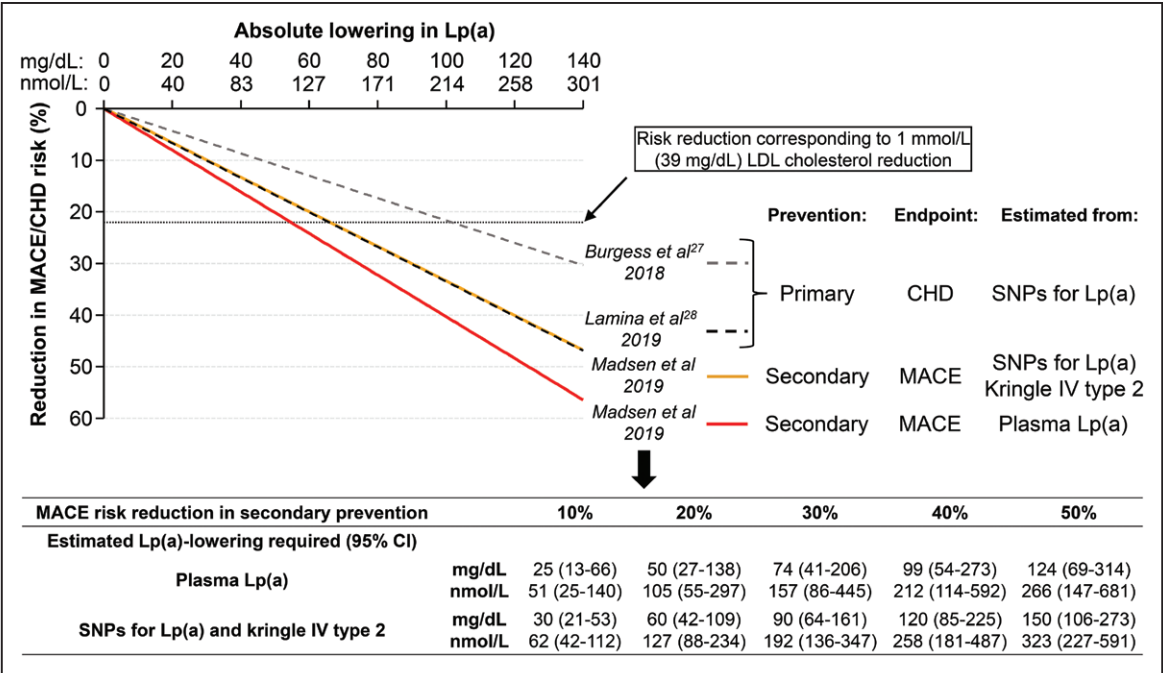


Figure 7. Estimated absolute Lp(a) (lipoprotein[a])-lowering needed for major adverse cardiovascular event (MACE)/coronary heart disease risk reduction.

The solid red line indicates the estimate based on plasma Lp(a) in 2527 individuals from the Copenhagen General Population Study with a median follow-up of 5 y from a multifactorially adjusted Cox proportional hazards regression. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides. The solid orange line indicates the estimate based on single-nucleotide polymorphisms for Lp(a) and kringle IV type 2 in 2131 individuals from the Copenhagen General Population Study from instrumental variable analysis scaled to a short-term clinical trial setting. The dashed lines indicate the predicted primary prevention trial estimates from Burgess et al²⁷ (gray) and Lamina et al²⁸ (black) for the end point coronary heart disease. The horizontal dashed line indicates the 22% major vascular event risk reduction per 1 mmol/L (39 mg/dL) reduction in LDL (low-density lipoprotein) cholesterol observed in combined primary and secondary prevention statin outcome trials from the 2010 coronary heart disease Cholesterol Treatment Trialists' Collaboration meta-analysis.²⁶

a secondary prevention setting. The current data on the possible role of Lp(a) in secondary prevention primarily stem from post hoc analyses of randomized clinical trials including individuals with coronary artery disease, where recruitment did not target individuals with high Lp(a), and evaluating therapies not aimed specifically at lowering Lp(a).¹²⁻¹⁵ These analyses have not surprisingly yielded mixed results with high between-study heterogeneity.^{12, 14, 16} A recent nested case-cohort analysis of the dal-Outcomes randomized clinical trial including individuals with recent acute coronary syndrome did not find any association between Lp(a) and risk of recurrent CVD events¹³ nor did a study of individuals with coronary heart disease, focusing on mortality only.³² This is however in contrast to a recent meta-analysis of placebo-controlled randomized statin outcome trials, which combined both primary and secondary prevention populations, and reported that individuals with Lp(a)≥50 mg/dL had an age- and sex-adjusted hazard ratio for CVD events of 1.31 (95% CI, 1.08–1.58) as compared with individuals with Lp(a)<15 mg/dL.¹⁴ Post hoc analyses of the FOURIER trial supported this with an adjusted hazard ratio for CVD events of 1.22 (95% CI, 1.01–1.48) for the highest versus the lowest quartile of Lp(a).

These previous analyses have either focused on quantiles of Lp(a) or did not include cut points of Lp(a) higher than 50 mg/dL, highlighting, as also pointed out by the authors, a lack of data on the risk of recurrent CVD events in individuals with high Lp(a) concentrations.¹⁴ Data on risk of CVD in individuals with Lp(a) above 75 mg/dL or 100 mg/dL are of interest, as future cardiovascular outcome trials would likely include patients with high Lp(a). Studies using single nucleotide polymorphisms for Lp(a) have estimated that a reduction in Lp(a) of 60 or 100 mg/dL short-term (ie, 5 years) could be needed to obtain results comparable to a 1 mmol/L (39 mg/dL) reduction in LDL cholesterol in primary prevention and for the end point coronary heart disease.^{27, 28} In contrast, we have in this study estimated that a reduction in Lp(a) of ≈50 mg/dL (105 nmol/L) would be needed to obtain a similar risk reduction for MACE in a secondary prevention setting. Of note though, the precise absolute reduction in Lp(a) needed to obtain a clinically meaningful reduction in risk of CVD would of course be dependent on primary versus secondary prevention and on the specific assay used, as available Lp(a) assays currently are not uniformly calibrated and standardized; indeed, the study yielding the estimate of 100 mg/dL included a multitude of different

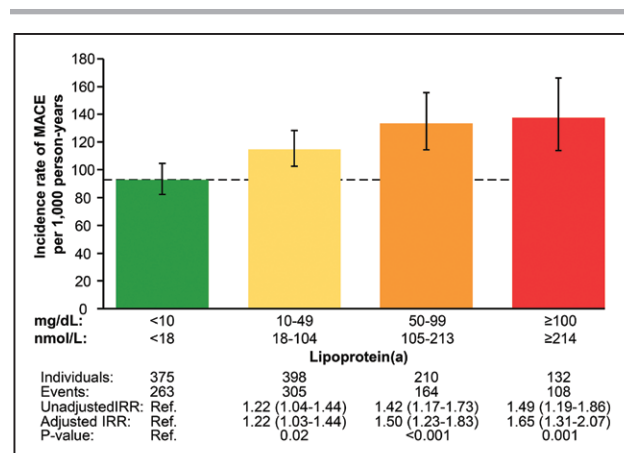


Figure 8. Absolute risk of major adverse cardiovascular event (MACE) according to concentrations of Lp(a) (lipoprotein[a]) in the Copenhagen City Heart Study and Copenhagen Ischemic Heart Disease Study combined.

Based on 1115 individuals from the 2 studies combined. Incidence rates and incidence rate ratios with 95% CIs are from Poisson regression. Error bars indicate 95% CIs. Multifactorial adjustment was for age, sex, lipid-lowering therapy, diabetes mellitus, hypertension, body mass index, smoking status, alcohol intake, physical activity, Lp(a)-corrected LDL cholesterol, and triglycerides. IRR indicates incidence rate ratio.

assays with variable absolute measurements.^{27, 31} In the present study, we provide absolute and relative risks of recurrent CVD events in individuals with Lp(a) ≥ 100 mg/dL using a commercially available standardized assay with minimal isoform bias and show that the risk increases stepwise already from levels < 50 mg/dL, and in those with Lp(a) ≥ 100 mg/dL a 2-fold higher risk of MACE was observed as compared with individuals with Lp(a) < 10 mg/dL.

Previous studies in both primary³³ and secondary¹² prevention have questioned whether Lp(a) is independently associated with CVD in individuals with low LDL or total cholesterol concentrations. In our study, results were similar for those with low Lp(a)-corrected LDL cholesterol concentrations or on lipid-lowering therapy, which is in accordance with the recent meta-analysis of primary and secondary prevention statin outcome trials.¹⁴ This could indicate that lowering Lp(a) would be of benefit, even in individuals who are well treated on standard lipid-lowering therapy. It is often neglected that conventional LDL cholesterol assays measure the cholesterol content in both LDL and Lp(a) particles.²⁵ This implies that in some individuals, the reason for suboptimal response to potent LDL-lowering therapy and failure to reach desired LDL cholesterol targets might be high Lp(a) concentrations.

Plasma levels of Lp(a) are highly genetically determined³⁴ and the genetic data in this study indicate a causal role of Lp(a) in recurrent CVD similar to incident CVD.³⁰ This is also supported by the current finding that risk associated with high Lp(a) was the highest in younger individuals, where the possible causal effect

of genetically determined Lp(a) is not masked by other strong cardiovascular risk factors accumulating with age. Nonetheless, confirmation of causality requires randomized cardiovascular outcome trials. To date, no randomized cardiovascular outcome trials targeting individuals with high Lp(a) with drugs specifically aimed at lowering Lp(a) have been conducted. However, antisense oligonucleotides targeting apolipoprotein(a) and lowering Lp(a) levels by 80% to 90% have concluded phase 2 development and are likely to proceed development in a phase 3 cardiovascular outcome trial.^{17, 18} Some drugs that are already approved for treatment of dyslipidemia are capable of reducing Lp(a) in plasma, but only to a limited extent as PCSK9 inhibitors and niacin can reduce Lp(a) 20% to 30%, whereas the effect of statins on Lp(a) seems negligible or even slightly increases Lp(a) levels.^{15, 17, 18, 31, 35}

Our study has limitations. First, we only included white individuals of Danish descent. Hence, we cannot determine if our results apply to other ethnicities such as individuals of African descent whom overall have higher concentrations of Lp(a) than white individuals;³¹ however, we are not aware of data suggesting that the present findings are not applicable to other ethnicities. Second, the absolute concentration of Lp(a) measured varies considerably according to the specific assay used due to the lack of uniform calibration and standardization of Lp(a) assays and possible measurement bias due to apolipoprotein(a) being heterogenous in size and number of antibody epitopes, 2 facts often neglected in studies on Lp(a). In the present study, we used different assays, but all measurements were calibrated to the standardized, and commercially available assay by Denka Seiken using fresh samples to facilitate comparison and reproduction of results in other cohorts. Also, it is unlikely that there was a systematic bias to use a particular assay more often within a given stratum of Lp(a). That being said, differences in the assays used, their calibration and the conversion from mg/dL to nmol/L as done in this study needs to be taken into consideration when interpreting the results. Third, as the main analyses included only individuals with previous CVD, this stratification could have introduced an index event or collider bias, which could have affected the observed risk estimates, including overestimation of the Lp(a)-lowering needed for a certain risk reduction.^{16, 36, 37} This form of bias could be present in both the observational and the genetic analyses. Fourth, we used nonfasting lipid profiles which could influence primarily the triglyceride measurements; however, the use of nonfasting lipid profiles are now widely endorsed by different scientific societies.^{20, 21} Fifth, we did not have information on the intensity of statin treatment in individuals from the studies, hence we do not know if they were on the maximum tolerated statin dose. As compared with previous studies done post hoc in randomized clinical trials, our study represents data from the

general population where use of preventive cardiovascular therapies varies and is often not fully in accordance with the standard of care proposed by current guidelines. This is both a limitation for the present study, but concomitantly a strength as data from population-based or real-world studies likely better reflecting the actual treatment outside of the more stringent and selective randomized clinical trials. Additional strengths include the use of the national Danish health registries ensuring excellent follow-up and confirmation of the results from the contemporary CGPS in older cohorts with higher baseline risks, as these individuals were included in a different time period with higher prevalence of cardiovascular risk factors and before the widespread use of lipid-lowering therapy. Last, it is important to remember that the estimates for required Lp(a)-lowering obtained in this study, as well as in previous studies, are subject to some degree of uncertainty. Especially for the genetic estimate, multiple choices and assumptions are made which could each shift the estimate toward a smaller or bigger lowering of Lp(a) needed for a particular risk reduction including type of assay, included instruments and their effect on the exposure Lp(a), primary or secondary prevention setting, choice of end point, index event bias, statistical assumptions, scaling from lifelong genetic exposure to short-term trial, etc. Specifically, our conversion of lifelong to short-term risk reduction rests on the assumptions that: (1) the fraction between lifelong and short-term risk reduction is similar for LDL cholesterol and Lp(a); (2) the fraction is similar for the end points MACE and coronary heart disease; (3) the fraction is similar for secondary and primary prevention. These are of course strong assumptions and would imply that the mechanisms by which Lp(a) cause CVD is similar to those of LDL cholesterol, but that is not necessarily the case.³⁰ All in all, the provided estimates from this and previous studies are only estimates, and it has yet to be shown that lowering of Lp(a) will result in reduced risk of CVD and if so, to what degree.

CONCLUSIONS

High concentrations of Lp(a) were associated with high risk of recurrent CVD events in individuals from the general population with preexisting CVD. To achieve a 20% MACE risk reduction in secondary prevention, we estimate that Lp(a) should be reduced 50 mg/dL (105 nmol/L) short-term (ie, 5 years).

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REFERENCES

1. Kamstrup PR, Benn M, Tybjaerg-Hansen A, Nordestgaard BG. Extreme lipoprotein(a) levels and risk of myocardial infarction in the general population: the Copenhagen city heart study. *Circulation*. 2008;117:176–184. doi: 10.1161/CIRCULATIONAHA.107.715698
2. Erqou S, Kaptoge S, Perry PL, Di Angelantonio E, Thompson A, White IR, Marcovina SM, Collins R, Thompson SG, Danesh J; Emerging Risk Factors Collaboration. Lipoprotein(a) concentration and the risk of coronary heart disease, stroke, and nonvascular mortality. *JAMA*. 2009;302:412–423. doi: 10.1001/jama.2009.1063
3. Kamstrup PR, Tybjaerg-Hansen A, Steffensen R, Nordestgaard BG. Genetically elevated lipoprotein(a) and increased risk of myocardial infarction. *JAMA*. 2009;301:2331–2339. doi: 10.1001/jama.2009.801
4. Clarke R, Peden JF, Hopewell JC, Kyriakou T, Goel A, Heath SC, Parish S, Barlera S, Franzosi MG, Rust S, et al; PROCARDIS Consortium. Genetic variants associated with Lp(a) lipoprotein level and coronary disease. *N Engl J Med*. 2009;361:2518–2528. doi: 10.1056/NEJMoa0902604
5. Thanassoulis G, Campbell CY, Owens DS, Smith JG, Smith AV, Peloso GM, Kerr KF, Pechlivanis S, Budoff MJ, Harris TB, et al; CHARGE Extracoronary Calcium Working Group. Genetic associations with valvular calcification and aortic stenosis. *N Engl J Med*. 2013;368:503–512. doi: 10.1056/NEJMoa1109034
6. Emdin CA, Khera AV, Natarajan P, Klarin D, Won HH, Peloso GM, Stitziel NO, Nomura A, Zekavat SM, Bick AG, et al; CHARGE–Heart Failure Consortium; CARDIOGRAM Exome Consortium. Phenotypic characterization of genetically lowered human lipoprotein(a) levels. *J Am Coll Cardiol*. 2016;68:2761–2772. doi: 10.1016/j.jacc.2016.10.033
7. Kamstrup PR, Nordestgaard BG. Elevated lipoprotein(a) levels, LPA risk genotypes, and increased risk of heart failure in the general population. *JACC Heart Fail*. 2016;4:78–87. doi: 10.1016/j.jchf.2015.08.006
8. Catapano AL, Graham I, De Backer G, Wiklund O, Chapman MJ, Drexel H, Hoes AW, Jennings CS, Landmesser U, Pedersen TR, et al; ESC Scientific Document Group. 2016 ESC/EAS guidelines for the management of dyslipidaemias. *Eur Heart J*. 2016;37:2999–3058. doi: 10.1093/eurheartj/ehw272
9. Grundy SM, Stone NJ, Bailey AL, Beam C, Birtcher KK, Blumenthal RS, Braun LT, de Ferranti S, Faiella-Tommasino J, Forman DE, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APHA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines. *Circulation*. 2019;139:e1082–e1143. doi: 10.1161/CIR.0000000000000625
10. Nordestgaard BG, Chapman MJ, Ray K, Borén J, Andreotti F, Watts GF, Ginsberg H, Amarencio P, Catapano A, Descamps OS, et al; European Atherosclerosis Society Consensus Panel. Lipoprotein(a) as a cardiovascular risk factor: current status. *Eur Heart J*. 2010;31:2844–2853. doi: 10.1093/eurheartj/ehq386
11. Davidson MH, Ballantyne CM, Jacobson TA, Bittner VA, Braun LT, Brown AS, Brown WV, Cromwell WC, Goldberg RB, McKenney JM, et al. Clinical utility

of inflammatory markers and advanced lipoprotein testing: advice from an expert panel of lipid specialists. *J Clin Lipidol*. 2011;5:338–367. doi: 10.1016/j.jacl.2011.07.005

12. O'Donoghue ML, Morrow DA, Tsimikas S, Sloan S, Ren AF, Hoffman EB, Desai NR, Solomon SD, Domanski M, Arai K, et al. Lipoprotein(a) for risk assessment in patients with established coronary artery disease. *J Am Coll Cardiol*. 2014;63:520–527. doi: 10.1016/j.jacc.2013.09.042
13. Schwartz GG, Ballantyne CM, Barter PJ, Kallend D, Leiter LA, Leitersdorf E, McMurray JJV, Nicholls SJ, Olsson AG, Shah PK, et al. Association of lipoprotein(a) with risk of recurrent ischemic events following acute coronary syndrome: analysis of the dal-outcomes randomized clinical trial. *JAMA Cardiol*. 2018;3:164–168. doi: 10.1001/jamacardio.2017.3833
14. Willeit P, Ridker PM, Nestel PJ, Simes J, Tonkin AM, Pedersen TR, Schwartz GG, Olsson AG, Colhoun HM, Kronenberg F, et al. Baseline and on-treatment lipoprotein(a) levels for prediction of cardiovascular events: individual patient-data meta-analysis of statin outcome trials. *Lancet*. 2018;392:1311–1320. doi: 10.1016/S0140-6736(18)31652-0
15. O'Donoghue ML, Fazio S, Giugliano RP, Stroes ESG, Kanevsky E, Gouni-Berthold I, Im K, Lira Pineda A, Wasserman SM, Češka R, et al. Lipoprotein(a), pcsk9 inhibition, and cardiovascular risk. *Circulation*. 2019;139:1483–1492. doi: 10.1161/CIRCULATIONAHA.118.037184
16. Boffa MB, Stranges S, Klar N, Moriarty PM, Watts GF, Koschinsky ML. Lipoprotein(a) and secondary prevention of atherothrombotic events: A critical appraisal. *J Clin Lipidol*. 2018;12:1358–1366. doi: 10.1016/j.jacl.2018.08.012
17. Viney NJ, van Capelleveen JC, Geary RS, Xia S, Tami JA, Yu RZ, Marcovina SM, Hughes SG, Graham MJ, Crooke RM, et al. Antisense oligonucleotides targeting apolipoprotein(a) in people with raised lipoprotein(a): two randomised, double-blind, placebo-controlled, dose-ranging trials. *Lancet*. 2016;388:2239–2253. doi: 10.1016/S0140-6736(16)31009-1
18. Tsimikas S, Karwatowska-Prokopcuk E, Gouni-Berthold I, Tardif JC, Baum SJ, Steinhagen-Thiessen E, Shapiro MD, Stroes ES, Moriarty PM, Nordestgaard BG, et al. Safety and efficacy of akcea-apo(a)-lrx to lower lipoprotein(a) levels in patients with established cardiovascular disease: A phase 2 dose-ranging trial. American Heart Association Scientific Sessions. 2018.
19. Langsted A, Kamstrup PR, Benn M, Tybjaerg-Hansen A, Nordestgaard BG. High lipoprotein(a) as a possible cause of clinical familial hypercholesterolaemia: a prospective cohort study. *Lancet Diabetes Endocrinol*. 2016;4:577–587. doi: 10.1016/S2213-8587(16)30042-0
20. Langlois MR, Chapman MJ, Cobbaert C, Mora S, Remaley AT, Ros E, Watts GF, Borén J, Baum H, Bruckert E, et al.; European Atherosclerosis Society (EAS) and the European Federation of Clinical Chemistry and Laboratory Medicine (EFLM) Joint Consensus Initiative. Quantifying atherogenic lipoproteins: current and future challenges in the era of personalized medicine and very low concentrations of LDL cholesterol. a consensus statement from EAS and EFLM. *Clin Chem*. 2018;64:1006–1033. doi: 10.1373/clinchem.2018.287037
21. Nordestgaard BG, Langsted A, Mora S, Kolovou G, Baum H, Bruckert E, Watts GF, Sypniewska G, Wiklund O, Borén J, et al.; European Atherosclerosis Society (EAS) and the European Federation of Clinical Chemistry and Laboratory Medicine (EFLM) joint consensus initiative. Fasting is not routinely required for determination of a lipid profile: clinical and laboratory implications including flagging at desirable concentration cut-points—a joint consensus statement from the European Atherosclerosis Society and European Federation of Clinical Chemistry and Laboratory Medicine. *Eur Heart J*. 2016;37:1944–1958. doi: 10.1093/eurheartj/ehw152
22. Kamstrup PR, Tybjaerg-Hansen A, Nordestgaard BG. Elevated lipoprotein(a) and risk of aortic valve stenosis in the general population. *J Am Coll Cardiol*. 2014;63:470–477. doi: 10.1016/j.jacc.2013.09.038
23. Langsted A, Varbo A, Kamstrup PR, Nordestgaard BG. Elevated Lipoprotein(a) does not cause low-grade inflammation despite causal association with aortic valve stenosis and myocardial infarction: a study of 100,578 individuals from the general population. *J Clin Endocrinol Metab*. 2015;100:2690–2699. doi: 10.1210/jc.2015-1096
24. Langsted A, Kamstrup PR, Nordestgaard BG. High lipoprotein(a) and high risk of mortality. *Eur Heart J*. 2018.
25. Yeang C, Witztum JL, Tsimikas S. 'LDL-C'=LDL-C+Lp(a)-C: implications of achieved ultra-low LDL-C levels in the proprotein convertase subtilisin/kexin type 9 era of potent LDL-C lowering. *Curr Opin Lipidol*. 2015;26:169–178. doi: 10.1097/MOL.0000000000000171
26. Baigent C, Blackwell L, Emberson J, Holland LE, Reith C, Bhalra N, Peto R, Barnes EH, Keech A, Simes J, et al.; Cholesterol Treatment Trialists' (CTT) Collaboration. Efficacy and safety of more intensive lowering of LDL cholesterol: a meta-analysis of data from 170,000 participants in 26 randomised trials. *Lancet*. 2010;376:1670–1681. doi: 10.1016/S0140-6736(10)61350-5
27. Burgess S, Ference BA, Staley JR, Freitag DF, Mason AM, Nielsen SF, Willeit P, Young R, Surendran P, Karthikeyan S, et al.; European Prospective Investigation Into Cancer and Nutrition–Cardiovascular Disease (EPIC-CVD) Consortium. Association of LPA variants with risk of coronary disease and the implications for lipoprotein(a)-lowering therapies: a Mendelian randomization analysis. *JAMA Cardiol*. 2018;3:619–627. doi: 10.1001/jamacardio.2018.1470
28. Lamina C, Kronenberg F; Lp(a)-GWAS-Consortium. Estimation of the required lipoprotein(a)-lowering therapeutic effect size for reduction in coronary heart disease outcomes: a Mendelian randomization analysis. *JAMA Cardiol*. 2019;4:575–579. doi: 10.1001/jamacardio.2019.1041
29. Ference BA, Ginsberg HN, Graham I, Ray KK, Packard CJ, Bruckert E, Hegele RA, Krauss RM, Raal FJ, Schunkert H, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J*. 2017;38:2459–2472. doi: 10.1093/eurheartj/ehx144
30. Nordestgaard BG, Langsted A. Lipoprotein (a) as a cause of cardiovascular disease: insights from epidemiology, genetics, and biology. *J Lipid Res*. 2016;57:1953–1975. doi: 10.1194/jlr.R071233
31. Tsimikas S. A test in context: lipoprotein(a): diagnosis, prognosis, controversies, and emerging therapies. *J Am Coll Cardiol*. 2017;69:692–711. doi: 10.1016/j.jacc.2016.11.042
32. Zewinger S, Kleber ME, Tragante V, McCubrey RO, Schmidt AF, Direk K, Laufs U, Werner C, Koenig W, Rothenbacher D, et al.; GENIUS-CHD consortium. Relations between lipoprotein(a) concentrations, LPA genetic variants, and the risk of mortality in patients with established coronary heart disease: a molecular and genetic association study. *Lancet Diabetes Endocrinol*. 2017;5:534–543. doi: 10.1016/S2213-8587(17)30096-7
33. Verbeek R, Hoogeveen RM, Langsted A, Stiekema LCA, Verweij SL, Hovingh GK, Wareham NJ, Khaw KT, Boekholdt SM, Nordestgaard BG, et al. Cardiovascular disease risk associated with elevated lipoprotein(a) attenuates at low low-density lipoprotein cholesterol levels in a primary prevention setting. *Eur Heart J*. 2018;39:2589–2596. doi: 10.1093/eurheartj/ehy334
34. Schmidt K, Noureen A, Kronenberg F, Utermann G. Structure, function, and genetics of lipoprotein (a). *J Lipid Res*. 2016;57:1339–1359. doi: 10.1194/jlr.R067314
35. Nordestgaard BG, Nicholls SJ, Langsted A, Ray KK, Tybjaerg-Hansen A. Advances in lipid-lowering therapy through gene-silencing technologies. *Nat Rev Cardiol*. 2018;15:261–272. doi: 10.1038/nrcardio.2018.3
36. Dahabreh IJ, Kent DM. Index event bias as an explanation for the paradoxes of recurrence risk research. *JAMA*. 2011;305:822–823. doi: 10.1001/jama.2011.163
37. Munafò MR, Tilling K, Taylor AE, Evans DM, Davey Smith G. Collider scope: when selection bias can substantially influence observed associations. *Int J Epidemiol*. 2018;47:226–235. doi: 10.1093/ije/dyx206